

Pinned distance densities in nested train track models

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Abstract

We study the original pinned distance probabilities of explicit deterministic planar product-digit measures with singular coordinate marginals. For every rational $1 < \alpha < 2$, a superlacunary construction has Hausdorff dimension α and packing dimension two. Its distance densities have the uniform attained exponent $q_* = (2 - \alpha)/(3 - 2\alpha)$ when $\alpha < 3/2$, and are bounded and jointly continuous when $\alpha \geq 3/2$. The finite exponent is sharp precisely at pins whose first coordinate belongs to the horizontal support; off that support the densities are locally bounded and continuous. At a fixed stage ratio the critical exponent changes and its finite endpoint fails. We determine the corresponding exceptional pin sets and their Hausdorff and packing dimensions. A two-step bounded-variation and coarea estimate gives a fixed positive restriction with a jointly continuous density at every pin, and hence intervals of uniformly positive length in all the studied distance sets. At the fixed-ratio transition the densities belong to every finite L^q , with an $O(\log q)$ norm upper bound and a double-exponential tail upper bound, but have no continuous version on the horizontal-support pin set. Essential boundedness there remains open. These conclusions use the exact rectangular placement and stage schedule; they are not general Falconer-type dimension thresholds.

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1 Introduction and main results

For a pin $p \in \mathbb{R}^2$, write $D_p(y) = |y - p|$. The size of the compact set $D_p(E)$ and the regularity of a particular probability $(D_p)_*\mu$ supported on it are different questions. This paper gives explicit nested examples in which every studied distance set contains an interval, while the original probability can fail L^2 at every original pin. It also shows how changing the stage schedule changes both the critical integrability exponent and whether that exponent is attained.

Throughout, $\alpha \in (1, 2)$ is rational, and

$$a = \frac{\alpha - 1}{2}, \quad b = \frac{\alpha}{2}, \quad c = 1 - b, \quad \beta = \alpha - 1. \quad (1.1)$$

In particular $b - a = 1/2$ and $a + c = 1/2$. Section 2 constructs compact sets E_- and E_+ and probabilities $\lambda = \kappa \times \zeta_-$ and $\mu = \kappa \times \zeta_+$ on them. For a fixed $\eta = 2^{-k} < 1/8$, they lie respectively in

$$P_- = [0, \eta]^2, \quad P_+ = [0, \eta] \times [1 - \eta, 1].$$

Pins range over the whole rectangle $P = P_-$. Set

$$H = \text{supp } \kappa, \quad \mathcal{B} = H \times [0, \eta], \quad \nu_p = (D_p)_*\mu.$$

Thus $E_- \subset \mathcal{B} \subset P$. When ν_p is absolutely continuous, g_p denotes its density. All L^q norms below use Lebesgue measure on $\mathbb{R}$. A continuous density means a continuous version on all of $\mathbb{R}$, including the endpoints of its support.

Theorem 1.1 (Superlacunary schedule). *Use the digit construction of Section 2 with $L_j/n_{j-1} \rightarrow \infty$. Then both $E_\pm$ have Hausdorff dimension α and packing dimension two, and both λ, μ are α -Frostman probabilities with singular coordinate marginals. Every ν_p , $p \in P$, is absolutely continuous.*

(i) If $1 < \alpha < 3/2$, put

$$q_* = \frac{2 - \alpha}{3 - 2\alpha} = \frac{c}{c - a}. \quad (1.2)$$

Then $\sup_{p \in P} \|g_p\|_{q_} < \infty$. For every $p \in \mathcal{B}$, the exact membership range is $1 \leq q \leq q_*$; no such density belongs to any L^q with $q > q_*$, including $q = \infty$.*

(ii) If $3/2 \leq \alpha < 2$, the densities admit one bounded jointly continuous version on $P \times \mathbb{R}$.

(iii) In every parameter range, the original density is locally jointly continuous on $(P \setminus \mathcal{B}) \times \mathbb{R}$, and $\|g_p\|_\infty \leq C / \text{dist}(p_1, H)$ for $p \notin \mathcal{B}$.

At $\alpha = 4/3$, the uniform upper conclusion is L^2 for all $p \in P$; failure of every $L^{2+\varepsilon}$ applies exactly to pins with $p_1 \in H$. There is no conflict with boundedness off H . The endpoint $\alpha = 3/2$ is a direct uniform-convergence theorem, not a formal limiting value of the finite exponent.

Theorem 1.2 (Fixed stage ratio). *Use $L_j = Kn_{j-1}$ for $j \geq 2$, with an integer $K \geq 1$, and set*

$$D_K = K(3/2 - \alpha) + 1 - \alpha, \quad \beta_K = \frac{\alpha + 2aK}{1 + aK}. \quad (1.3)$$

The sets $E_\pm$ have Hausdorff dimension α and packing dimension $\beta_K < 2$, and their probabilities are α -Frostman with singular coordinate marginals. Assume $aK > 2$ for the following regularity statements.

- (i) If $D_K > 0$, set $q_K = c(K + 1)/D_K$. For every $1 \leq q < q_K$, $\sup_{p \in P} \|g_p\|_q < \infty$. At every $p \in \mathcal{B}$, the exact range is $1 \leq q < q_K$: the finite critical endpoint fails.
- (ii) If $D_K < 0$, the original densities are bounded and jointly continuous on $P \times \mathbb{R}$.
- (iii) If $D_K = 0$, all finite L^q upper bounds hold, and

$$\sup_{p \in P} \|g_p\|_q \leq C \log(e + q), \quad 1 \leq q < \infty. \quad (1.4)$$

For each $p \in \mathcal{B}$ there is no continuous-density version. Whether $g_p \in L^\infty$ at such pins remains open. Equation (1.4) is only an upper bound; no matching norm growth lower bound is asserted.

- (iv) Off $\mathcal{B}$, the local bounded-continuity conclusion of Theorem 1.1(iii) holds in all three regimes.

Theorem 1.3 (Positive restrictions and actual distance intervals). *Assume either the superlacunary schedule, or the fixed-ratio schedule with $aK > 1$. For each fixed $0 < \varepsilon \leq 1$, the positive subprobability*

$$\nu_{p,\varepsilon} = (D_p)_*(\mathbf{1}_{\{|y_1 - p_1| \geq \varepsilon\}} \mu)$$

has a jointly continuous density $g_\varepsilon(p, r)$ satisfying

$$0 \leq g_\varepsilon(p, r) \leq C/\varepsilon, \quad \nu_{p,\varepsilon}(\mathbb{R}) \geq 1 - C\varepsilon^{\alpha-1}.$$

There exists $\ell > 0$ such that every $D_p(E_+)$, $p \in P_-$, contains an interval of length ℓ . The reversed assertion holds from P_+ to E_- . In particular every pin in $E_- \cup E_+$ has an interval in its distance set to that union. The interval may depend on the pin; only its length lower bound is uniform.

We also determine the exact exceptional sets for stronger density regularity, their dimensions, and positive-mask stability. At the fixed-ratio boundary we obtain a double-exponential upper tail and uniform double-exponential integrability on the bounded output interval.

1.1 Relation to earlier work and verification status

The near-track obstruction and far-track smoothing distinction already appears in Guth–Iosevich–Ou–Wang [1, Sections 1.2 and 6]. Their general good-source construction uses a wave-packet decomposition. Here the restriction is an actual positive, fixed horizontal-band deletion for one explicitly placed nested model. Neither that finite-scale intuition nor a general distance-set threshold is claimed as new. Peres–Schlag [2] provides classical generalized-projection background; no result from that theory is an input to the proofs below.

This candidate consolidates four analytic source notes after a separate model-based analytic review and a limited recheck of two scope clarifications. That process is not professional human peer review or formal verification. The consolidation is expository, and no priority claim or comprehensive literature-novelty certification is made. The package’s source map records the exact inputs and editorial changes. All mathematical arguments needed here are included in the paper.

2 The digit construction and dimensions

Choose positive integers L_j with aL_j , $L_j/2$, and bL_j integral. Let $n_0 = 0$ and $n_j = \sum_{i=1}^j L_i$. At stage j , the free horizontal binary positions, counted relative to n_{j-1} , are

$$\{1, \dots, aL_j\} \cup \{L_j/2 + 1, \dots, L_j\}; \quad (2.1)$$

the free vertical positions are $\{1, \dots, bL_j\}$. All remaining digits are zero. All free bits in both coordinates are independent and fair. Let κ_X and κ_Y be the resulting probabilities.

Fix k with $\eta = 2^{-k} < 1/8$ and take L_1 sufficiently large that the first k positions are free in both coordinates. Condition κ_X on its all-zero length- k prefix to obtain κ . Condition κ_Y on its all-zero and all-one length- k prefixes to obtain ζ_- and ζ_+ respectively. Define

$$\lambda = \kappa \times \zeta_-, \quad \mu = \kappa \times \zeta_+, \quad E_{\pm} = \text{supp}(\kappa \times \zeta_{\pm}). \quad (2.2)$$

Supports are images of the compact coding spaces. At a binary endpoint one uses a supporting closed coding cylinder; no zero-mass endpoint is added by an arbitrary half-open convention. The fixed prefix conditioning changes mass constants by 2^k in each coordinate, and never changes a later stage exponent.

For every $p \in P_-$ and $y \in P_+$, including filled-tail points, set

$$v_0 = 1 - 2\eta > 0, \quad D_0 = 2; \quad y_2 - p_2 \geq v_0, \quad |y - p| \leq D_0. \quad (2.3)$$

Hence the vertical derivative of distance is at least v_0/D_0 . All output probabilities are supported in $I = [v_0, D_0]$.

2.1 Two exact schedules

In the superlacunary schedule, assume

$$\frac{L_j}{n_{j-1}} \rightarrow \infty \quad (j \geq 2). \quad (2.4)$$

For any rational α , choose an integer d for which $ad, bd, d/2$ are integers, choose an admissible large L_1 , and take $L_j = d(n_{j-1} + 1)^2$ for $j \geq 2$. This gives an actual realization. For example $\alpha = 13/10$, $\eta = 1/16$, $L_1 = 40$, and $L_j = 20(n_{j-1} + 1)^2$ give $a = 3/20$, $b = 13/20$ and $n_1 - aL_2 = -5003$.

In the fixed-ratio schedule, an integer $K \geq 1$ is fixed and

$$L_j = Kn_{j-1}, \quad n_j = (K + 1)n_{j-1} \quad (j \geq 2). \quad (2.5)$$

Choosing L_1 divisible by the necessary denominators preserves all integrality requirements. In particular $n_{j-1} = n_1(K + 1)^{j-2}$. No rounding argument or assertion for irrational α is used.

The original non- L^2 example was restricted to $1 < \alpha < 4/3$ and also imposed $(2 - 3\alpha/2)L_j - \alpha n_{j-1} \rightarrow \infty$. Condition (2.4) implies that condition in this range. There is no need to impose it for the enlarged family $1 < \alpha < 2$.

2.2 Frostman bounds and Hausdorff dimension

Let $F_X(N), F_Y(N)$ count free positions through depth N , before prefix conditioning. In a stage of length L the relative counts are

$$f_X(l) = \min(l, aL) + \max(0, l - L/2), \quad f_Y(l) = \min(l, bL).$$

Their sum equals

$$f_X(l) + f_Y(l) = \begin{cases} 2l, & 0 \leq l \leq aL, \\ l + aL, & aL \leq l \leq L/2, \\ 2l - cL, & L/2 \leq l \leq bL, \\ l + (\alpha - 1)L, & bL \leq l \leq L. \end{cases} \quad (2.6)$$

Each expression is at least αl , and the endpoint value is αL . Consequently

$$F_X(N) + F_Y(N) \geq \alpha N, \quad F_X(n_j) = F_Y(n_j) = bn_j. \quad (2.7)$$

For $N \geq k$, every occupied depth- N square has mass $2^{2k-F_X(N)-F_Y(N)}$. A ball of radius comparable to 2^{-N} meets only a bounded number of such squares. Thus both probabilities obey the α -Frostman estimate $\rho(B(x, r)) \leq Cr^\alpha$. At $N = n_j$ there are at most $C2^{\alpha n_j}$ occupied squares, so $\dim_{\text{H}} E_{\pm} \leq \alpha$. The Frostman estimate gives the reverse inequality and hence equality.

Also $f_X(l) \geq \beta l$ and $b \geq \beta$, whence

$$F_X(N) \geq \beta N, \quad \kappa([u - \varepsilon, u + \varepsilon]) \leq K_X \varepsilon^\beta \quad (0 < \varepsilon \leq 1). \quad (2.8)$$

The interval bound follows by comparison with dyadic intervals and includes all real u . In particular κ is nonatomic. Each coordinate support has a cover at depth n_j by $C2^{bn_j}$ intervals of length 2^{-n_j} . The total length tends to zero because $b < 1$. The coordinate laws are therefore singular, and are nonatomic because their free counts diverge. There is no hidden Lebesgue coordinate in the limiting product.

2.3 Packing dimensions of the planar supports

Under (2.4), in each sufficiently late stage the first aL_j bits of both coordinates are free. Every fixed coding cylinder thus contains at least 2^{2aL_j} occupied squares at depth $n_{j-1} + aL_j$, and

$$\frac{2aL_j}{n_{j-1} + aL_j} \rightarrow 2.$$

Every such cylinder has upper box dimension two. Every nonempty relatively open subset of the support contains a coding cylinder. To pass from upper box to packing dimension, use the countable-cover characterization by modified upper box dimension: take relative closures in any countable cover, which do not change upper box dimension. Baire's theorem implies that one closure contains a relatively open subset. Its upper box dimension is two, proving $\dim_{\text{P}} E_{\pm} = 2$.

For (2.5), the total-count ratio during a stage is $[\alpha n + f_X(l) + f_Y(l)]/(n + l)$, with $L = Kn$. It is monotone on each of the four pieces in (2.6). Its values at $l = 0, L/2, L$ are α , and its two possible maxima are

$$B_1 = \frac{\alpha + 2aK}{1 + aK}, \quad B_2 = \frac{\alpha + (3b - 1)K}{1 + bK}.$$

After clearing denominators, the numerator of $B_1 - B_2$ is $a(1 - b)K^2 > 0$. Thus the upper box dimension is $\beta_K = B_1 < 2$. A fixed coding prefix changes the free count by a constant, so every cylinder has this same upper box dimension. The preceding Baire argument proves

$$\dim_{\text{H}} E_{\pm} = \alpha, \quad \dim_{\text{P}} E_{\pm} = \begin{cases} 2, & \text{superlacunary,} \\ (\alpha + 2aK)/(1 + aK), & \text{fixed ratio.} \end{cases} \quad (2.9)$$

No regular subset is substituted for the original supports.

2.4 Filled tails and the exact replacement

Choose a stage j_0 with $n_{j_0} \geq k$. For $j \geq j_0$, define μ_j by retaining the first n_j digits of μ and replacing all later digits by independent fair bits. Equivalently, each occupied 2^{-n_j} -square is filled uniformly with its exact μ -mass. Then $\mu_j \rightarrow \mu$.

Inside a parent square Q of side $h = 2^{-n}$, where $n = n_{j-1}$, the conditional parent law is uniform and the next law is the affine image of $\xi_{A,w} \times \zeta_{B,v}$, where

$$\begin{aligned}\xi_{A,w} &= \frac{1}{A} \sum_{i=0}^{A-1} \text{Unif}[i/A, i/A + w], \\ \zeta_{B,v} &= \frac{1}{B} \sum_{l=0}^{B-1} \text{Unif}[l/B, l/B + v], \\ A &= 2^{aL}, \quad B = 2^{bL}, \quad w = 2^{-L/2}, \quad v = 2^{-L}, \quad L = L_j.\end{aligned}\tag{2.10}$$

The initial horizontal free block chooses the A positions; the last half of the horizontal bits and the filled tail produce the entire width w . The vertical free block chooses B positions and the filled tail supplies width v . Therefore

$$Bw = A, \quad Bv = 2^{-cL}, \quad w \leq A^{-1}, \quad v \leq B^{-1}.\tag{2.11}$$

This exact placement, rather than the number of descendants alone, is the input to the next section.

3 Two bounded variation quadrature estimates

We first isolate the deterministic geometric estimate used at every stage. Let U denote uniform probability on $[0, 1]$.

Lemma 3.1 (Equal-bin quadrature). *If a probability ξ assigns mass $1/N$ to each of the N equal bins of $[0, 1]$, then for every function F of bounded variation,*

$$\left| \int F d\xi - \int F dU \right| \leq \frac{\text{Var}(F)}{N}.\tag{3.1}$$

Proof. Within each bin the difference between the two conditional averages is bounded by the oscillation of F . Summing with weight $1/N$ proves the claim, since the sum of the bin oscillations is at most total variation. At jump endpoints use one-sided representatives consistent with the bin convention. The template laws (2.10) used below are absolutely continuous within bins, so endpoint values do not affect their integrals. $\square$

Let $Q = [q_1, q_1 + h] \times [q_2, q_2 + h]$, $0 < h \leq 1$, satisfy (2.3) with respect to the pin p . Let U_Q be uniform probability on Q , and let ρ_Q be the affine image of the product in (2.10), allowing any integers $A, B \geq 1$ and widths $0 < w \leq A^{-1}$, $0 < v \leq B^{-1}$. Write

$$M_\varepsilon(p, y) = \mathbf{1}_{\{|y_1 - p_1| \geq \varepsilon\}}.$$

When a norm of a signed pushforward is written below, it means the norm of its density. Each constituent measure is positive and unnormalized.

Lemma 3.2 (Fixed-band comparison). *Both masked distance pushforwards have densities, and*

$$\|(D_p)_*(M_\varepsilon \rho_Q) - (D_p)_*(M_\varepsilon U_Q)\|_\infty \leq \frac{4D_0}{h} \left(\frac{1}{v_0 A} + \frac{1}{\varepsilon B w} \right).\tag{3.2}$$

The pin and the mask are identical in the two terms.

Proof. Use coordinates $X = q_1 + hu$, $Y = q_2 + hz$ and the local output $t = |y - p|/h$. Conversion back to physical radius contributes a final factor h^{-1} to the density.

First fix a horizontal strip $i/A \leq u \leq i/A + w$, keep the part $|X - p_1| \geq \varepsilon$, and average in the vertical coordinate. For a fixed local output t , set $r = ht$. Horizontal coarea gives the t -density

$$G_i(z, t) = \frac{1}{w} \sum_{\text{allowed roots } X} \frac{r}{|X - p_1|}, \quad X - p_1 = \pm \sqrt{r^2 - (Y - p_2)^2}. \quad (3.3)$$

Indeed $\partial t / \partial u = (X - p_1)/r$, so no extra factor of h belongs in (3.3). On each branch, $Y - p_2 > 0$ and $|X - p_1|$ decreases as z increases. The reciprocal derivative $r/|X - p_1|$ is monotone. Requiring the root to lie in the strip cuts out an interval of z , and imposing the band cutoff only shortens it. The zero extension of each monotone interval function has variation at most twice its supremum. There are at most two branches, each at most $D_0/(\varepsilon w)$, so

$$\text{Var}_z G_i(z, t) \leq \frac{4D_0}{\varepsilon w}. \quad (3.4)$$

Lemma 3.1 replaces $\zeta_{B,v}$ by U at a density cost $4D_0/(\varepsilon Bw)$. Averaging the strips with their $1/A$ weights preserves that bound. The width v has not entered except through $v \leq 1/B$.

With the vertical coordinate uniform, fix u . There is only one allowed vertical root because $Y - p_2 \geq v_0$. Its local-output density is

$$J(u, t) = \mathbf{1}_{\{\text{allowed vertical root}\}} \mathbf{1}_{\{|X - p_1| \geq \varepsilon\}} \frac{r}{Y - p_2}. \quad (3.5)$$

It is at most D_0/v_0 . On either side of $X = p_1$ it is monotone in $|X - p_1|$; requiring the root to lie in the vertical side of Q leaves at most one interval per side. Therefore $\text{Var}_u J(u, t) \leq 4D_0/v_0$. A second application of Lemma 3.1, this time to $\xi_{A,w}$, costs $4D_0/(v_0 A)$. Adding the two errors and returning to physical radius proves the claim. Coarea is applied on strictly monotone intervals; the cutoff removes the only horizontal critical point. All bounds hold for almost every output, which is sufficient for the stated essential supremum. $\square$

Lemma 3.3 (Harmonic improvement). *At the particular physical cutoff $\varepsilon = h/A$,*

$$\|(D_p)_*(M_{h/A} \rho_Q) - (D_p)_*(M_{h/A} U_Q)\|_\infty \leq CD_0 \left(\frac{1}{hv_0 A} + \frac{\log(2A)}{h^2 Bw} \right). \quad (3.6)$$

Proof. For the physical strip $I_i = [q_1 + hi/A, q_1 + hi/A + hw]$, let d_i be the infimum of $|X - p_1|$ over its nonempty retained part. Empty strips are omitted. Then $d_i \geq h/A$, and the sharper variation bound in the first part of the preceding proof is $4D_0/(wd_i)$. Each strip lies in its own coarse bin of length h/A . Order bins to the left and right of p_1 . Except for at most a fixed number of nearest neighbors, the l -th bin has distance at least $lh/(2A)$. The cutoff controls the nearest bins. The same ordering works if p_1 lies outside Q . Hence

$$\frac{1}{A} \sum_i d_i^{-1} \leq \frac{C}{h} \left(1 + \sum_{l=1}^A \frac{1}{l} \right) \leq \frac{C}{h} \log(2A). \quad (3.7)$$

Average this sharper variation over strips before the vertical quadrature step. Its local-output error is at most $CD_0 \log(2A)/(hBw)$; physical scaling gives the second term in (3.6). The horizontal quadrature term is unchanged. $\square$

The resolving relation $Bw = A \rightarrow \infty$ expresses a vertical sampling spacing fine enough to resolve an actual horizontal interval. A Frostman bound or a numerical branching profile does not imply either lemma.

4 Positive filters and intervals at every pin

The arguments in this section apply whenever

$$S = \sum_{j>j_0} 2^{n_{j-1}-aL_j} < \infty. \quad (4.1)$$

This follows from superlacunarity, since eventually $aL_j \geq 2n_{j-1}$ and $n_{j-1} \geq j-1$. For a fixed ratio it follows from $aK > 1$, since the summands are $2^{-(aK-1)n_{j-1}}$.

Proposition 4.1 (One fixed positive restriction). *For every $p \in P$ and fixed $0 < \varepsilon \leq 1$, the actual measure $\nu_{p,\varepsilon} = (D_p)_*(M_\varepsilon(p, \cdot)\mu)$ has a jointly continuous density $g_\varepsilon(p, r)$, with*

$$\|g_\varepsilon(p, \cdot)\|_\infty \leq C_*/\varepsilon, \quad \nu_{p,\varepsilon}(\mathbb{R}) \geq 1 - K_X \varepsilon^\beta. \quad (4.2)$$

One possible constant is

$$C_* = (D_0/v_0)2^{n_{j_0}} + 4D_0(1 + v_0^{-1})S. \quad (4.3)$$

It may depend on the finitely many initial stages, but is independent of p and any finest observation scale.

Proof. Let $g_{j,p,\varepsilon}$ be the masked density for μ_j . Apply Lemma 3.2 in every parent square, multiply by its exact parent mass, and add. The masses sum to one. By $Bw = A$,

$$\|g_{j,p,\varepsilon} - g_{j-1,p,\varepsilon}\|_\infty \leq 4D_0(v_0^{-1} + \varepsilon^{-1})2^{n_{j-1}-aL_j}. \quad (4.4)$$

No count of cells appears. At the initial filled stage, vertical coarea on each square gives $\|g_{j_0,p,\varepsilon}\|_\infty \leq (D_0/v_0)2^{n_{j_0}}$. Thus the densities converge in L^∞ uniformly in p , and the bound (4.3) follows.

For a fixed p , the discontinuities of the mask are on $y_1 = p_1 \pm \varepsilon$. These lines have zero μ -mass by (2.8). Weak convergence can therefore be applied to a bounded continuous distance test function multiplied by the mask. It identifies the weak limit with $(D_p)_*(M_\varepsilon\mu)$. The common bounded output support makes the L^∞ convergence also L^1 convergence, identifying its density with the same measure. This argument works for *each* pin with the same bounds, without an exceptional pin set. The retained-mass estimate follows directly from (2.8).

To obtain the continuous version, write the finite filled-tail density f_j as a finite linear combination of rectangle indicators. Polar coarea gives, for $r > 0$,

$$g_{j,\varepsilon}(p, r) = r \int_0^{2\pi} f_j(p + r\omega(\theta)) \mathbf{1}_{\{|r \cos \theta| \geq \varepsilon\}} d\theta, \quad \omega(\theta) = (\cos \theta, \sin \theta). \quad (4.5)$$

A circle with positive radius meets each rectangle boundary in only finitely many angles. The mask equality also has at most finitely many angles. For a convergent sequence of (p, r) the integrand therefore converges at almost every angle and has a bounded integrable majorant. Dominated convergence proves joint continuity. Around $r = 0$ the density is zero by uniform vertical separation; extend by zero for negative r .

For continuous radius fibers the supremum is the essential supremum. Thus (4.4) is a uniform supremum estimate on the continuous versions over $P \times \mathbb{R}$. Their uniform limit is jointly continuous and represents the actual positive restriction at every pin. $\square$

Corollary 4.2 (Raw absolute continuity and weak integrability). *Every ν_p is absolutely continuous. Its density satisfies*

$$\sup_{p \in P} |\{r : g_p(r) > T\}| \leq CT^{-\alpha} \quad (T > 0). \quad (4.6)$$

In particular $\sup_p \|g_p\|_q < \infty$ for $1 \leq q < \alpha$.

Proof. As $\varepsilon \downarrow 0$, the positive restricted measures increase to ν_p , since $\mu\{y_1 = p_1\} = 0$. A Lebesgue-null set has zero mass for every restriction and hence for the limit. This proves absolute continuity. Write $g_p = g_\varepsilon(p, \cdot) + b_{p,\varepsilon}$ with $b_{p,\varepsilon} \geq 0$ and $\int b_{p,\varepsilon} \leq K_X \varepsilon^\beta$. For sufficiently large T , choose $\varepsilon = 2C_*/T \leq 1$. Then the good density is at most $T/2$, so Markov's inequality gives

$$|\{g_p > T\}| \leq \frac{2}{T} \int b_{p,\varepsilon} \leq 2K_X(2C_*)^\beta T^{-(1+\beta)}.$$

Since $1 + \beta = \alpha$, this is the claim for large T ; bounded output support handles smaller T . Layer-cake integration gives the stated strong L^q bounds. $\square$

Proposition 4.3 (Uniform length in the actual distance sets). *There is $\ell > 0$ such that every $D_p(E_+)$, $p \in P_-$, contains an interval of length ℓ . The reversed conclusion holds for $p \in P_+$ and source E_- .*

Proof. Choose a fixed $\varepsilon_0 > 0$ with $K_X \varepsilon_0^\beta \leq 1/2$. Every nonnegative fiber of g_{ε_0} then has mass at least $1/2$ on the interval $I = [v_0, D_0]$ of length $W = D_0 - v_0$. It has some point r_p with $g_{\varepsilon_0}(p, r_p) \geq 1/(2W)$. Joint continuity and compactness, using the continuous zero extension outside I , supply one $\delta > 0$ such that

$$|r - r_p| \leq \delta \implies g_{\varepsilon_0}(p, r) \geq 1/(4W)$$

for every p . The support of this submeasure is contained in the compact set $D_p(E_+)$ because the submeasure is a positive restriction of the actual source law. Thus the entire interval is in the actual image, not only in a finite approximation or an almost-everywhere version of that image. Taking any fixed $\ell \leq \delta$ proves the first assertion.

In the reverse configuration the vertical separation has negative sign, but the absolute derivative bound is unchanged. The vertical branch is still unique and the BV monotonicities simply reverse sign. The same proof therefore applies. If necessary take the smaller of the two length bounds. $\square$

Proposition 4.3 proves Theorem 1.3. In particular, failure of L^2 or even unboundedness of the raw density is compatible with an interval in every one of the distance images here.

5 The attained superlacunary endpoint

We now use the exact global product structure to strengthen Corollary 4.2. Write $g_{j,p}$ for the unmasked filled-tail distance density. In a stage beginning at $n = n_{j-1}$, write $L = L_j$, $h = 2^{-n}$ and use the parameters in (2.10). Decompose the *signed increment* $g_{j,p} - g_{j-1,p}$ using the auxiliary band

$$\varepsilon_j = h/A. \tag{5.1}$$

This changing band is only a proof device; it is not the fixed positive restriction of Section 4.

5.1 The outer and inner parts of a stage increment

Lemma 3.3 and $Bw = A$ give, after weighting by parent masses, an outer-part bound

$$\|\text{outer increment}\|_\infty \leq C(1 + L)2^{2n-aL}. \tag{5.2}$$

It also bounds every finite L^q norm up to a constant depending only on I . Under superlacunarity the series is summable: eventually $\log_2(1 + L) \leq aL/2$ and $aL/2 \geq 3n$, leaving $C2^{-n}$.

For the inner part it is essential to estimate the *global* horizontal mass. At depth n , every occupied horizontal prefix interval has κ -mass 2^{k-bn} . The next-stage coarse bins of width h/A are globally aligned; each occupied strip has mass $2^{k-bn}/A$. A band of length $2h/A$ meets only a bounded number of these bins. For both the new filled law and the old filled law this gives

$$M_j, M_{j-1} \leq C_k A^{-1} 2^{-bn}. \quad (5.3)$$

For the old law this is also immediate from its piecewise constant horizontal density and the band's length. The corresponding vertical densities satisfy

$$H_j = 2^{k+c(n+L)}, \quad H_{j-1} = 2^{k+cn}. \quad (5.4)$$

Horizontal restriction does not change the vertical conditional law: both measures are actual product measures.

For a product submeasure of horizontal mass M with vertical density at most H , vertical coarea gives a distance-density supremum at most $(D_0/v_0)MH$. Its total mass is M . Hence, putting $\sigma = 1 - 1/q$, with $1/\infty = 0$,

$$\|\text{band density}\|_q \leq CMH^\sigma, \quad 1 \leq q \leq \infty. \quad (5.5)$$

For $q = \infty$ this is the direct supremum bound. Substituting (5.3)–(5.4) gives for the new law

$$C2^{[-b+c\sigma]n}2^{[-a+c\sigma]L}. \quad (5.6)$$

The old law has the additional favorable factor $2^{-c\sigma L}$. The norm of their difference is bounded by the sum of these positive norms.

Proposition 5.1 (Uniform endpoint upper bound). *If $1 < \alpha < 3/2$, then $\sup_{p \in P} \|g_p\|_{q_*} < \infty$ for q_* in (1.2). If $3/2 \leq \alpha < 2$, then the original densities are uniformly bounded and have a jointly continuous version on $P \times \mathbb{R}$.*

Proof. For $\alpha < 3/2$, the endpoint identity is

$$c(1 - 1/q_*) = a.$$

Thus (5.6) reduces exactly to $C2^{-(b-a)n} = C2^{-n/2}$. For $\alpha \geq 3/2$, take $q = \infty$, so $\sigma = 1$ and the same bound becomes

$$C2^{-(\alpha-1)n}2^{-(\alpha-3/2)L} \leq C2^{-n/2}. \quad (5.7)$$

In either case the inner parts are summable. Together with (5.2), this shows

$$\sup_{p \in P} \|g_{j,p} - g_{j-1,p}\|_q \leq C[(1 + L_j)2^{2n_{j-1}-aL_j} + 2^{-n_{j-1}/2}], \quad (5.8)$$

for the stated endpoint q . The initial filled density is bounded. Therefore the unmasked stage densities are uniformly Cauchy in L^q . Their weak limits are the actual ν_p , since D_p is continuous and $\mu_j \rightharpoonup \mu$. On the common bounded output interval, L^q convergence implies L^1 convergence and identifies the density. No discontinuous mask remains in this final identification.

For $q = \infty$, the unmasked finite densities are jointly continuous by (4.5) with the band indicator removed. The continuous versions satisfy the same supremum estimates and converge uniformly. The limit is jointly continuous. In particular the case $\alpha = 3/2$ follows from the explicit summable bound $C2^{-n/2}$ in (5.7). $\square$

5.2 The track lower bound for every horizontal-support pin

Let $p \in \mathcal{B}$. At stage j select a supporting horizontal coding interval of depth $n + L/2$ containing p_1 . Its width is $2^{-n-L/2}$ and its exact horizontal mass is

$$m_j = 2^{k-bn-aL}. \quad (5.9)$$

This remains valid at coding endpoints by selecting one supporting closed cylinder. The vertical support has at most $2^{b(n+L)-k}$ occupied intervals of length $\delta_j = 2^{-(n+L)}$. On the horizontal track,

$$0 \leq |y - p| - (y_2 - p_2) = \frac{(y_1 - p_1)^2}{|y - p| + (y_2 - p_2)} \leq \frac{2^{-2n-L}}{2v_0} \leq C\delta_j.$$

Its distance image is therefore contained in a Borel set U_j with

$$|U_j| \leq C2^{-c(n+L)-k}, \quad \nu_p(U_j) \geq m_j. \quad (5.10)$$

The argument uses only $p_1 \in H$ and the vertical separation, not membership of p_2 in the lower vertical Cantor support.

For every finite $q > 1$, Hölder's inequality on this particular set gives

$$\int_{U_j} g_p(r)^q dr \geq \frac{m_j^q}{|U_j|^{q-1}} \geq c_q 2^{[q(c-a)-c]L + [q(c-b)-c]n}. \quad (5.11)$$

The inequality also holds with an infinite left side. When $1 < \alpha < 3/2$ and $q > q_*$, the coefficient of L is positive; superlacunarity makes the right side diverge. Hence $g_p \notin L^q$. Compact output support then also excludes L^∞ . At $q = q_*$ the residual exponent is instead $-q_*n/2$, consistent with the attained upper endpoint. This proves the sharp part of Theorem 1.1; off- H behavior is proved in Section 7.

Consequently the superlacunary regimes on $\mathcal{B}$ are: $1 < \alpha < 4/3$ gives an attained $q_* < 2$ and failure of L^2 ; $\alpha = 4/3$ gives L^2 and failure of all $L^{2+\varepsilon}$; $4/3 < \alpha < 3/2$ gives an attained finite $q_* > 2$; and $\alpha \geq 3/2$ gives bounded jointly continuous densities on the entire pin rectangle. These nonmembership statements never extend to off- H pins.

6 Fixed ratio and the excluded critical endpoint

Assume (2.5). The fixed-mask proof and interval theorem already hold when $aK > 1$. In this section use the convenient common hypothesis $aK > 2$. Set

$$D_K = K(c - a) + (c - b) = K(3/2 - \alpha) + 1 - \alpha. \quad (6.1)$$

If $D_K > 0$, define $q_K = c(K + 1)/D_K$. Notice that

$$c(K + 1) - D_K = b + aK > 0,$$

so $q_K > 1$ as required for the Hölder argument.

6.1 Upper estimates

The outer estimate (5.2) becomes

$$C(1 + Kn)2^{-(aK-2)n}, \quad (6.2)$$

which is summable since $aK > 2$ and $n = n_1(K+1)^{j-2}$. The global inner estimate (5.6) becomes

$$C2^{[-b-aK+c(K+1)(1-1/q)]n} = C2^{[D_K-c(K+1)/q]n}. \quad (6.3)$$

Its constants arise from the same aligned-bin mass bound, true product structure, and vertical coarea estimate; the old filled law has the additional favorable factor $2^{-c(1-1/q)L}$.

If $D_K > 0$, the exponent in (6.3) is negative exactly when $q < q_K$. If $D_K < 0$, it is negative for $q = \infty$. If $D_K = 0$, it is negative for every fixed finite q . Combining the inner and outer bounds proves uniform L^q Cauchy convergence in precisely the upper ranges asserted in Theorem 1.2. Weak convergence identifies the limiting original law. In the L^∞ case the continuous finite versions converge uniformly, yielding joint continuity.

For a particular finite exponent one can weaken the outer condition. Every conditional output from a parent square lies in an interval of length at most $\sqrt{2}h$, because distance is 1-Lipschitz. Apply this support-length gain before summing conditional increments with their parent masses. Minkowski's inequality gives the outer L^q estimate

$$C_q(1 + Kn)2^{(2-1/q-aK)n}. \quad (6.4)$$

Thus sufficient conditions for a given finite q are

$$aK > 2 - 1/q, \quad D_K - c(K+1)/q < 0. \quad (6.5)$$

If $D_K > 0$, the whole subcritical range follows under the weaker strict condition $aK > 2 - 1/q_K$. These are sufficient conditions furnished by the proof, not claims that $aK = 1$ or $aK = 2$ is an exact geometric threshold.

6.2 Failure at the endpoint

For every $p \in \mathcal{B}$, the track image (5.10) satisfies $|U_j| \rightarrow 0$. Substituting $L = Kn$ into (5.11) gives

$$\int_{U_j} g_p(r)^q dr \geq c_q 2^{[qD_K - c(K+1)]n}. \quad (6.6)$$

For $q > q_K$ the right side diverges. More importantly, for $q = q_K$ it is bounded below by a fixed positive constant. If $g_p \in L^{q_K}$, the absolute continuity of the Lebesgue integral of $g_p^{q_K}$ would force the left side to tend to zero as $|U_j| \rightarrow 0$, a contradiction. The U_j need not be disjoint or nested. A constant lower bound for the *whole* norm alone would not prove endpoint failure; localization to shrinking-measure sets is essential. This proves the exact open range on $\mathcal{B}$.

6.3 Examples and the endpoint switch

For $\alpha = 13/10$, $a = 3/20$ and $b = 13/20$. The fixed ratio $K = 14$ satisfies $aK = 21/10 > 2$ and gives

$$D_K = 5/2, \quad q_K = 21/10, \quad \beta_K = 55/31.$$

All pins in P have L^2 original densities, and pins in $\mathcal{B}$ have exactly $1 \leq q < 21/10$. For the superlacunary construction with the same α , $q_* = 7/4$ is attained, every pin in $\mathcal{B}$ fails L^2 , and the planar packing dimension is two. These are two specific schedules with controlled placements; the comparison does not show that packing dimension alone determines the phase diagram.

The boundary $D_K = 0$ occurs at

$$\alpha = \frac{3}{2} - \frac{1}{2(K+1)}. \quad (6.7)$$

For example $K = 9$, $\alpha = 29/20$ gives $aK = 81/40 > 2$. The next section proves noncontinuity on $\mathcal{B}$ and the logarithmic norm upper bound there, while keeping the L^∞ question open.

The endpoint contrast has a precise exponent explanation. In the superlacunary regime the leading critical L exponent cancels while the inherited horizontal mass leaves the summable $2^{-n/2}$ gain. At a fixed ratio the shifted critical exponent cancels the entire n -and- L power simultaneously in (6.6), leaving a positive amount of L^{qK} mass on the shrinking track images.

7 Exact exceptional pin sets and the boundary

7.1 Off the horizontal support

Proposition 7.1. *Under the superlacunary assumptions, or the fixed-ratio assumptions with $aK > 1$, every $p \in P \setminus \mathcal{B}$ has a bounded continuous original density. The density is jointly continuous locally on $(P \setminus \mathcal{B}) \times \mathbb{R}$, and*

$$\|g_p\|_\infty \leq \frac{C}{\text{dist}(p_1, H)}. \quad (7.1)$$

Proof. Since H is compact, $d = \text{dist}(p_1, H) > 0$. Set $\varepsilon = d/2$. On a neighborhood of p with $\text{dist}(p'_1, H) > \varepsilon$, the mask $M_\varepsilon(p', \cdot)$ is identically one on the entire source support. The jointly continuous positive-filter density of Proposition 4.1 is therefore the original density throughout that neighborhood, with bound $2C_*/d$. $\square$

Every fixed off- H pin has a positive gap, so no third membership class arises from a rate of approximation to H . Bounds may diverge as pins approach H , which is a quantitative issue distinct from membership.

7.2 Dimensions of the full horizontal-support pin set

Proposition 7.2. *In the superlacunary regime,*

$$\dim_H H = \alpha - 1, \quad \dim_P H = 1, \quad \dim_H \mathcal{B} = \alpha, \quad \dim_P \mathcal{B} = 2. \quad (7.2)$$

In the fixed-ratio regime, put

$$s_K = \frac{b + aK}{1 + K/2} = \frac{\alpha + (\alpha - 1)K}{K + 2}, \quad (7.3)$$

$$t_K = \frac{b + aK}{1 + aK} = \frac{\alpha + (\alpha - 1)K}{2 + (\alpha - 1)K}. \quad (7.4)$$

Then

$$\dim_H H = s_K, \quad \dim_P H = t_K, \quad \dim_H \mathcal{B} = 1 + s_K, \quad \dim_P \mathcal{B} = 1 + t_K. \quad (7.5)$$

Proof. In the superlacunary case, (2.8) gives the Hausdorff lower bound for H . At depth $n + L/2$, the horizontal free count is $bn + aL$, whose ratio to depth tends to $2a = \alpha - 1$. The corresponding covers give the upper bound. At depth $n + aL$ the ratio tends to one, and every fixed cylinder has upper box dimension one. The coding-cylinder and Baire argument of Section 2 then gives packing dimension one.

In the fixed-ratio case the ratio $[bn + f_X(l)]/(n + l)$ increases up to $l = aKn$, decreases up to $l = Kn/2$, and increases thereafter. Its trough is s_K and its peak is t_K ; indeed $s_K \leq b \leq t_K$, so no stage endpoint has been omitted. The finitely many initial depths alter only a constant. The horizontal law is s_K -Frostman, and covers at the middle-stage trough give the matching

Hausdorff upper bound. Every fixed cylinder has upper box dimension t_K , and Baire gives the packing lower bound.

For $\mathcal{B} = H \times [0, \eta]$, multiply the appropriate horizontal Frostman law by Lebesgue probability on the interval to obtain the Hausdorff lower bounds. Multiply the trough covers by interval grids to obtain the upper bounds. For packing dimension, cylinder products with nondegenerate subintervals have upper box dimensions larger by exactly one; the same countable-cover/Baire argument applies. This proves the product formulas directly rather than relying on an equality for arbitrary fractal products. $\square$

Useful identities for this family are

$$D_K = (K + 2)(1/2 - s_K), \quad q_K = \frac{1 - s_K}{1 - 2s_K} \quad (D_K > 0). \quad (7.6)$$

The superlacunary formula has the same expression in $s = \alpha - 1$, although the endpoint behavior differs. At $D_K = 0$,

$$s_K = 1/2, \quad t_K = \frac{K + 1}{K + 2}, \quad \dim_H \mathcal{B} = 3/2, \quad \dim_P \mathcal{B} = 2 - \frac{1}{K + 2}. \quad (7.7)$$

For $K = 9$, $\alpha = 29/20$, these dimensions are $3/2$ and $21/11$. Under the standing requirement $aK > 2$, an integer boundary example must have $K \geq 9$.

7.3 No continuous version at the fixed-ratio boundary

Proposition 7.3. *If $aK > 2$ and $D_K = 0$, then every $p \in \mathcal{B}$ has no continuous original distance density on $\mathbb{R}$.*

Proof. The minimum vertical source coordinate is $r_0 = 1 - \eta$. For $p_1 \in H$, the point (p_1, r_0) lies in the source support, so

$$r_{\min}(p) = r_0 - p_2 > 0$$

is the minimum source distance. Let $\delta_j = 2^{-(n+L)}$. Combine the horizontal track in (5.9) with the lowest occupied vertical interval of length δ_j , which has ζ_+ -mass $2^{k-b(n+L)}$. The product subset has mass

$$2^{2k-2bn-(a+b)L}. \quad (7.8)$$

Its vertical distances are within δ_j of $r_{\min}(p)$ and its horizontal quadratic error is at most $C2^{-2n-L} \leq C\delta_j$. The actual distance image lies in $[r_{\min}(p), r_{\min}(p) + C_0\delta_j]$. Dividing (7.8) by δ_j gives

$$2^{2k}2^{(1-\alpha)n+(3/2-\alpha)L} = 2^{2k}2^{D_K n}. \quad (7.9)$$

At $D_K = 0$ this is a fixed positive constant, so

$$\nu_p([r_{\min}(p), r_{\min}(p) + C_0\delta_j]) \geq c_0\delta_j. \quad (7.10)$$

A continuous density for ν_p must vanish to the left of its support and hence at $r_{\min}(p)$. Its supremum on the shrinking displayed interval would tend to zero, giving mass $o(\delta_j)$, contrary to (7.10). $\square$

This proof does not exclude an essentially bounded density with a jump at the support endpoint. It settles continuity, not L^∞ membership.

7.4 The exact classification

Define

$$\begin{aligned}\text{Bad}_q &= \{p \in P : \nu_p \text{ has no } L^q \text{ density}\}, \\ \text{Bad}_{C^0} &= \{p \in P : \nu_p \text{ has no continuous density on } \mathbb{R}\}.\end{aligned}$$

All these laws are absolutely continuous. Compact support implies that every continuous-density version is bounded. The upper bounds, (5.11), (6.6), Proposition 7.1, and Proposition 7.3 therefore give the following exact statements.

1. Superlacunary, $1 < \alpha < 3/2$:

$$\text{Bad}_q = \emptyset \ (1 \leq q \leq q_*), \quad \text{Bad}_q = \mathcal{B} \ (q_* < q \leq \infty), \quad \text{Bad}_{C^0} = \mathcal{B}.$$

2. Superlacunary, $3/2 \leq \alpha < 2$: all Bad_q , $1 \leq q \leq \infty$, and Bad_{C^0} are empty.

3. Fixed ratio, $aK > 2$ and $D_K > 0$:

$$\text{Bad}_q = \emptyset \ (1 \leq q < q_K), \quad \text{Bad}_q = \mathcal{B} \ (q_K \leq q \leq \infty), \quad \text{Bad}_{C^0} = \mathcal{B}.$$

4. Fixed ratio, $aK > 2$ and $D_K < 0$: all these exceptional sets are empty.

5. Fixed ratio, $aK > 2$ and $D_K = 0$:

$$\text{Bad}_q = \emptyset \ (1 \leq q < \infty), \quad \text{Bad}_{C^0} = \mathcal{B}, \quad \text{Bad}_\infty \subseteq \mathcal{B}.$$

The exact set Bad_∞ in the last line is not determined here.

The nonempty exact exceptional sets have the dimensions in Proposition 7.2. They are exceptions to stronger *density regularity*, not pins with zero-measure or interval-free distance sets. Every pin in P still has the interval conclusion.

7.5 An upper bound on all boundary norms

Theorem 7.4. *Assume $aK > 2$ and $D_K = 0$. Then for a fixed $C < \infty$,*

$$\sup_{p \in P} \|g_p\|_q \leq C \log(e + q) \quad (1 \leq q < \infty). \quad (7.11)$$

There are $c_1, c_2, T_0 > 0$ such that

$$\sup_{p \in P} |\{r : g_p(r) > T\}| \leq \exp[-c_1 \exp(c_2 T)] \quad (T \geq T_0). \quad (7.12)$$

For some $c_3 > 0$,

$$\sup_{p \in P} \int_I \exp(\exp(c_3 g_p(r))) \, dr < \infty. \quad (7.13)$$

Only upper bounds are asserted; no matching lower growth in q or tail asymptotic is proved.

Proof. The initial density and the summable outer increments (6.2) have L^q bounds independent of $q \geq 1$, using the fixed finite output interval. The constant in (5.5) can also be chosen uniformly in q : the coarea factor is $(D_0/v_0)^{1-1/q}$ and fixed prefix factors are bounded by their $q = \infty$ values. At $D_K = 0$, the inner increment is at most $C2^{-c(K+1)n/q}$. Thus

$$\|g_p\|_q \leq C + C \sum_{j \geq 2} 2^{-c(K+1)n_1(K+1)^{j-2}/q}. \quad (7.14)$$

There are $O(1 + \log q)$ terms before the exponent is of order one; after that point the remaining sum is bounded by a fixed convergent series of the form $\sum_{l \geq 0} \exp[-A(K+1)^l]$. This proves (7.11).

For large T choose the real moment order $q = \exp[T/(4C)]$, increasing C and the lower threshold if necessary. Then $C \log(e + q) \leq T/2$ and Markov gives

$$|\{g_p > T\}| \leq (\|g_p\|_q/T)^q \leq 2^{-q},$$

which proves (7.12). Choose $0 < c_3 < c_2$ sufficiently small. Layer-cake integration against the derivative of $\exp(\exp(c_3 T))$ gives (7.13), since the negative term $-c_1 \exp(c_2 T)$ dominates $\exp(c_3 T)$ and the remaining linear terms. The integral is over I , not all of $\mathbb{R}$, so its nonzero constant baseline causes no divergence. $\square$

The double-exponential upper tail neither proves nor disproves essential boundedness. This completes all assertions of Theorem 1.2 and the full pin classification, with the stated boundary question intact.

8 Positive masks and nonuniform replacement stability

The fixed restriction controls arbitrary further positive source masks without requiring cancellation, a product structure for those masks, or an independent copy of the pin.

8.1 Domination and the retained mass cost

Fix $\varepsilon > 0$, and let $0 \leq F(p, y) \leq M_\varepsilon(p, y)$ be jointly Borel. Set

$$\sigma_p = (D_p)_*(F(p, \cdot)\mu), \quad m(p) = \int F(p, y) d\mu(y).$$

Positive domination gives a density f_p satisfying

$$0 \leq f_p \leq g_\varepsilon(p, \cdot) \leq C_*/\varepsilon. \quad (8.1)$$

Consequently

$$\|f_p\|_2^2 \leq (C_*/\varepsilon)m(p). \quad (8.2)$$

For $m(p) > 0$, normalization yields

$$\|f_p/m(p)\|_\infty \leq \frac{C_*}{\varepsilon m(p)}, \quad \|f_p/m(p)\|_2^2 \leq \frac{C_*}{\varepsilon m(p)}. \quad (8.3)$$

The squared L^2 norm is divided by $m(p)^2$; the unnormalized bound already contains one factor $m(p)$. Uniform normalized bounds therefore require a genuine lower bound on retained mass. One may integrate these fiber inequalities against any probability on P , using that same pin in both distance factors. Jointly measurable versions follow from the finite coarea formulas and convergence in the separable space $L^1(I)$.

Whenever the original density itself is uniformly bounded, as in the superlacunary range $\alpha \geq 3/2$ and the fixed-ratio range $D_K < 0$ under the stated growth hypothesis, the same argument applies to every $0 \leq F \leq 1$ without a preliminary band, using the raw supremum bound.

8.2 Previously imposed decreasing restrictions

Suppose $0 \leq \Gamma_N \leq 1$ is a decreasing sequence and

$$\int \Gamma_\infty d(\lambda \times \mu) \geq c_* > 0.$$

Choose one ε with $K_X \varepsilon^\beta < c_*/2$ and put $F_N = M_\varepsilon \Gamma_N$. This is still a decreasing family, loses at most $K_X \varepsilon^\beta$ additional pair mass, and has limiting pair mass at least $c_*/2$. Every fiber has the density bound (8.1); the limiting family has nonzero fibers on a positive- λ set of pins. All conditions encoded by Γ_N remain imposed.

This multiplication need not preserve a previously asserted relative retained fraction in every spatial cell. Such local fractions must be checked separately or obtained by further pruning with their actual mass budget. Positive domination survives any further pruning but does not supply its missing mass.

8.3 Endpoint perturbations and whole low frequency bounds

For a positive measure τ on $\mathbb{R}$, define

$$\text{Coll}_\delta(\tau) = \iint \mathbf{1}_{\{|s-t| \leq \delta\}} d\tau(s) d\tau(t).$$

Move a retained source endpoint by at most $A_0\delta$, possibly through a pin-dependent probability kernel. Pull the retained coupling back to its original marginal $F(p, y)\mu$ with $F \leq M_\varepsilon$ and mass $m(p)$. If two perturbed distances differ by at most δ , their original distances differ by at most $(1 + 2A_0)\delta$. Since the original positive density is bounded by C_*/ε , the perturbed law $\tilde{\sigma}_p$ satisfies

$$\text{Coll}_\delta(\tilde{\sigma}_p) \leq 2(1 + 2A_0)\delta (C_*/\varepsilon)m(p). \quad (8.4)$$

After division by $m(p)^2$, the probability version is bounded by $2(1 + 2A_0)\delta C_*/[\varepsilon m(p)]$. A shared pin displacement of magnitude at most $A_p\delta$ changes the factor $1 + 2A_0$ to $1 + 2A_0 + 2A_p$. The same physical pin is used before and after each two-distance comparison.

For completeness, with Fourier convention $\hat{\tau}(\xi) = \int e^{-2\pi i \xi t} d\tau(t)$, positivity gives

$$\int_{-R}^R |\hat{\tau}(\xi)|^2 d\xi \leq CR \text{Coll}_{1/R}(\tau). \quad (8.5)$$

Indeed take $k_\delta(t) = (1 - |t|/\delta)_+$ with $\delta = (2R)^{-1}$. Its Fourier transform is $\delta[\sin(\pi\delta\xi)/(\pi\delta\xi)]^2$, nonnegative and bounded below by a positive multiple of δ on $[-R, R]$. Fourier inversion gives

$$\int \hat{k}_\delta(\xi) |\hat{\tau}(\xi)|^2 d\xi = \iint k_\delta(s - t) d\tau(s) d\tau(t) \leq \text{Coll}_{1/R}(\tau),$$

which proves (8.5). Finite-measure Fourier inversion is legitimate since $\hat{k}_\delta$ is integrable. Thus (8.4) supplies whole-low-pass control. It does not imply a decaying high-frequency shell estimate for arbitrary varying masks or endpoint perturbations.

8.4 A weighted nonuniform sufficient condition

The positive-filter argument has a broader deterministic form that does not require a globally product limiting source. Its conclusion is weaker than the exact endpoint phase diagrams.

Theorem 8.1 (Weighted rectangular replacements). *Let ρ_j be probabilities on finite unions of filled dyadic squares in the separated source rectangle, converging weakly to ρ . At step j , replace each parent square Q , of side h_Q and mass m_Q , by the affine product template of*

Section 3, preserving its mass. Allow the parameters A_Q, B_Q, w_Q, v_Q and masses to depend on Q . Suppose the new filled leaves refine their parents, their maximum diameter tends to zero, and

$$\mathcal{E} = \sum_j \sum_{Q \text{ at step } j} \frac{m_Q}{h_Q} \left(A_Q^{-1} + (B_Q w_Q)^{-1} \right) < \infty. \quad (8.6)$$

Assume the first-coordinate marginal of ρ has no atoms. Then for each fixed $\varepsilon > 0$ and every pin in the separated pin rectangle, $(D_p)_*(M_\varepsilon \rho)$ has a density bounded by $C(1 + \varepsilon^{-1})$. Every original $(D_p)_* \rho$ is absolutely continuous. If the horizontal marginal satisfies a γ -Frostman interval bound, the raw densities are uniformly weak- $L^{1+\gamma}$.

Proof. Apply (3.2) with each parent's parameters, multiply by m_Q , and sum. Condition (8.6) makes the density increments absolutely summable in L^∞ . The initial finite mixture of filled squares has a bounded distance density by vertical coarea. Nonatomicity makes both hard-cut boundary lines ρ -null, so the same weak-limit identification as in Proposition 4.1 applies. Increasing the positive restrictions as $\varepsilon \downarrow 0$ gives raw absolute continuity. Finally the lost mass is $O(\varepsilon^\gamma)$; the Markov decomposition in Corollary 4.2 yields the weak- $L^{1+\gamma}$ estimate. $\square$

For uniform stages (8.6) reduces to

$$\sum_j 2^{n_j-1} [A_j^{-1} + (B_j w_j)^{-1}] < \infty.$$

The criterion records actual spatial resolution and accumulated weighted error. It covers adaptive parent-dependent templates with a common coordinate orientation, not arbitrary rotations. It does not by itself imply the exact critical exponents: those also use the globally aligned bin masses and product marginals in (5.3)–(5.4).

9 Scope and remaining questions

All conclusions concern the deterministic, rational-parameter digit models specified here and their original natural measures. The actual rectangular placement, the resolving relation $Bw = A$, and the schedule are substantive hypotheses. Neither the Frostman exponent, the packing dimension, nor a branch-count profile alone supplies them. The proof uses elementary bounded variation, coarea in a strictly monotone coordinate, weak and dominated convergence, interpolation, and Hölder and Markov inequalities. No general projection theorem, random-martingale theorem, or unrestricted recursive common-pin estimate is an input.

Fixing a sufficiently small ε gives an arbitrarily high retained fraction with a fixed good-law bound. Sending $\varepsilon \downarrow 0$ restores the original law, but the available bound C_*/ε diverges. In the superlacunary range $1 < \alpha < 4/3$, uniform L^2 restoration at a pin in $\mathcal{B}$ would contradict the proved non- L^2 original law: an L^2 -bounded sequence has a weak L^2 limit representing any weakly convergent measure limit. The auxiliary changing bands used to estimate signed increments were never assigned such a uniform L^2 bound.

The following matters remain outside the proved conclusions.

1. At $D_K = 0$ and $p_1 \in H$, essential boundedness and the exact set Bad_∞ remain unresolved. The noncontinuity argument allows a bounded jump, and the norm upper bound does not force unboundedness.
2. No matching lower bound for the $O(\log q)$ boundary norms or sharp double-exponential tail asymptotic is proved. At the finite fixed-ratio endpoint, weak- L^{q_K} or a sharper Lorentz classification is not determined.

3. The inequalities $aK > 1$ and $aK > 2$ are sufficient conditions for the displayed estimates. Equality cases, smaller ratios, and an exact necessary growth threshold are not settled.
4. No loss-free extension to arbitrary irrational α , arbitrary rotated templates, or exact nonproduct endpoints is asserted. Arbitrary source masks inserted *before* a replacement comparison need not preserve its geometry; positive domination applies to masks of the actual fixed good law after its construction.
5. Local mass retention for unrelated pre-existing pruning procedures must be checked with their own budgets. The estimates do not establish an unrestricted common-pin recursion or arbitrary-mask high-frequency shell decay.

The classical finite-scale insight is attributed to [1]; the model-specific conclusions here are not advertised as a new general Falconer threshold. A broad novelty assessment, professional peer review, and formal proof verification are not supplied by this manuscript.

A A direct subcritical increment argument

The endpoint proof in Section 5 uses a harmonic gain and global product masses. We retain here the alternative strictly subcritical argument, which uses a wider auxiliary band and estimates inside each parent square. Assume $1 < \alpha < 3/2$ and fix $1 < q < q_*$. Set

$$\sigma = 1 - 1/q, \quad \theta = \frac{1 + c\sigma/a}{2}, \quad \gamma_q = \frac{a - c\sigma}{2} > 0. \quad (\text{A.1})$$

Then $0 < \theta < 1$. In a parent square of side $h = 2^{-n}$, take the physical band width $\varepsilon_j = hd$ where $d = A^{-\theta}$. This is again a decomposition of a signed increment, not a good-law restoration with a uniform L^2 bound.

By Lemma 3.2 and $Bw = A$, the outer conditional L^∞ error is at most

$$Ch^{-1}[A^{-1} + (hdA)^{-1}] \leq Ch^{-2}A^{-(1-\theta)}.$$

The output lies in an interval of length $\sqrt{2}h$, so its L^q error is at most

$$C_q h^{-(2-1/q)} A^{-(1-\theta)}. \quad (\text{A.2})$$

Inside the band, the replacement has mass at most $2d + 2/A \leq 4d$, since $d \geq 1/A$ and the horizontal law assigns mass $1/A$ to each bin. The filled parent has band mass at most $2d$. The replacement's vertical relative-coordinate density is at most $(Bv)^{-1}$; physical vertical coarea gives an inner density supremum at most $Ch^{-1}d(Bv)^{-1}$. Its mass is at most $4d$, so interpolation yields

$$\|\text{inner conditional density}\|_q \leq C_q h^{-\sigma} d (Bv)^{-\sigma}. \quad (\text{A.3})$$

The parent obeys the same bound with Bv replaced by one and is therefore no larger. These finite-stage density estimates do not presume a Lebesgue vertical coordinate in the limiting singular law.

Using $Bv = 2^{-cL}$, combine (A.2) and (A.3). The conditional unmasked increment is at most

$$C_q [h^{-(2-1/q)} 2^{-a(1-\theta)L} + h^{-\sigma} 2^{-(a\theta - c\sigma)L}] \leq C_q 2^{(2-1/q)n - \gamma_q L}, \quad (\text{A.4})$$

since both L exponents equal γ_q by (A.1). Weighting by parent masses and applying Minkowski leaves this bound unchanged. Superlacunarity makes the series summable; for example, eventually $\gamma_q L_j \geq (3 - 1/q)n_{j-1}$, leaving $C_q 2^{-n_{j-1}}$. The initial density is bounded and weak convergence identifies the L^q limit with the actual original law. This recovers the strictly subcritical upper range directly. The separate harmonic and global-band argument is what retains the finite endpoint and proves the bounded endpoint at $\alpha = 3/2$.

References

- [1] L. Guth, A. Iosevich, Y. Ou, and H. Wang, *On Falconer's distance set problem in the plane*, arXiv:1808.09346v1 (2018), especially Sections 1.2 and 6. <https://arxiv.org/abs/1808.09346v1>.
- [2] Y. Peres and W. Schlag, *Smoothness of projections, Bernoulli convolutions, and the dimension of exceptions*, Duke Mathematical Journal **102** (2000), 193–251. <https://doi.org/10.1215/S0012-7094-00-10222-0>.